ROBOT CAR PROJECT · SHOPPING LIST

Everything to buy for one car

Phone-controlled 4WD car · ESP32-CAM · rear parking radar
BUY = quantity to buy (includes spares) · **Used** = on the car

Prices and stock from dzduino.com, checked 9 Nov 2026. The **ESP32-CAM, ESP32-CAM-MB, HC-SR04 and 2×18650 holder** showed **out of stock online**. Ask at the counter: the shop may still have them. **No LM2596 / buck converter is needed**: the Arduino makes the 5 V.

A • Main parts

✓	#	Part · what to ask for	Used	BUY	dzduino price	Note
☐	1	**4WD robot car chassis kit** 2 acrylic plates + **4 TT gear motors 3–6 V** + 4 wheels + screws/spacers	1	**1**	**4,200 DA** 4WD Smart Robot Car Chassis Kits with Speed Encoder **In Stock**	Often includes the 2×18650 holder and switch: check the box
☐	2	**ESP32-CAM** **AI-Thinker**, **OV2640** camera, 4 MB PSRAM	1	**1**	**3,200 DA** ESP32-CAM Module **Out of Stock online**	Camera + WiFi
☐	3	**ESP32-CAM-MB programmer** Micro-USB board the ESP32-CAM plugs into	1	**1**	**500 DA** ESP32-CAM-MB Micro USB Module (or USB-C version) **Out of Stock online**	For uploading code. Often sold together with the ESP32-CAM
☐	4	**Arduino UNO R3** ATmega328P, CH340 clone is fine, **with USB cable**	1	**1**	**2,300 DA** Arduino UNO Rev3-DIP (Compatible without logo) **In Stock**	Motors, sensor, buzzer
☐	5	**L298N motor driver** Red dual H-bridge board with heatsink	1	**1**	**1,000 DA** L298N Dual H Bridge DC Stepper Motor Driver Board **In Stock**	
☐	6	**HC-SR04 ultrasonic sensor** 5 V, 2–400 cm	1	**1**	**500 DA** Ultrasonic Ranging Sensor HC-SR04 **Out of Stock online**	Mounted at the rear
☐	7	**Buzzer, ACTIVE 5 V** Ask for "active" (beeps by itself). 2-pin or 3-pin module	1	**2**	**90 DA** 5V Active Buzzer **In Stock**	Not passive!
☐	8	**18650 Li-ion battery** 3.7 V, **real** 2000–3000 mAh, high-drain	2	**2**	**Not online: ask at the counter**	Avoid "9900 mAh" fakes
☐	9	**2×18650 battery holder** **Series** (7.4 V), with wires	1	**1**	**180 DA** 2x 18650 Battery holder without DC connector **Out of Stock online**	Skip if in the kit
☐	10	**18650 charger** 2-slot Li-ion charger, or 2 × TP4056 modules	1	**1**	**350 DA** Type-C 5V 1A 18650 TP4056 Charger Module (1 cell each) **In Stock**	Skip if you have one
☐	11	**On/off switch** Rocker or slide switch, ≥ 3 A	1	**1**	**Not online: ask at the counter**	Skip if in the kit

B • Small components

✓	#	Part · what to ask for	Used	BUY	dzduino price	Note
☐	12	**LED red 5 mm** Diffused red	1	**5**	**7 DA** LED 5mm (Red) · price each **In Stock**	Rear flashing LED, A0
☐	13	**LED white 5 mm** Bright white	3	**6**	**7 DA** LED 5mm (White) · price each **In Stock**	Reverse light (A2) + **2 headlights** (A4)
☐	14	**Resistor 220 Ω** 1/4 W	4	**10**	**5 DA** Resistor 1/4W · price each **In Stock**	One per LED (4 LEDs)

✓	#	Part · what to ask for	Used	BUY	dzduino price	Note
☐	15	**Resistor 1 kΩ** 1/4 W	1	**5**	**5 DA** Resistor 1/4W · price each **In Stock**	Voltage divider
☐	16	**Resistor 2 kΩ** 1/4 W (or use 2 × 1 kΩ in series)	1	**5**	**5 DA** Resistor 1/4W · price each **In Stock**	Voltage divider
☐	17	**Capacitor 2200 µF 16 V** Electrolytic (1000–2200 µF, 10 V or more also OK)	1	**1**	**Not online: ask at the counter**	At ESP32-CAM 5V: stops reboots
☐	18	**Capacitor 100 nF ("104")** Ceramic	4	**10**	**Not online: ask at the counter**	One across each motor

C · Wires, breadboard & mounting

✓	#	Part · what to ask for	Used	BUY	dzduino price	Note
☐	19	**Breadboard, half size** 400 points, with + / – rails both sides	1	**1**	**350 DA** 400 point breadboard 8.5 x 5.5cm **In Stock**	Holds the 5 V rails, LEDs, buzzer, divider
☐	20	**Jumper wires male-female** Dupont, 20 cm, pack of 40	18	**40**	**Not online: ask at the counter**	UNO ↔ L298N, sensor, ESP32-CAM, headlights at the front
☐	21	**Jumper wires male-male** Dupont, 20 cm, pack of 40	14	**40**	**Not online: ask at the counter**	UNO ↔ breadboard, breadboard links
☐	22	**Jumper wires female-female** Dupont, 20 cm, pack of 40	0	**40**	**400 DA** 1 Pin Dual-Female Jumper Wire 200mm (40 pcs pack) **In Stock**	Spare / FTDI wiring
☐	23	**Wire 22 AWG (0.5 mm²)** Red + black	1 m each	**2 m**	**Not online: ask at the counter**	Battery → switch → L298N, L298N → UNO VIN, motors
☐	24	**Heat-shrink tube** Assorted sizes	–	**1 set**	**60 DA** Heat Shrink Tubing (per metre, from) **In Stock**	Covers solder joints
☐	25	**Cable ties (zip ties)** Small, 100 mm	10	**1 pack**	**600 DA** CABLE TIES 8X200 100pcs **In Stock**	Tidy wiring
☐	26	**Double-sided foam tape** Strong	–	**1 roll**	**Not online: ask at the counter**	Mount boards on the chassis
☐	27	**M3 nylon standoffs + screws** 10 mm	8	**1 set**	**Not online: ask at the counter**	Mount UNO / L298N (optional)

Total of the parts with an online price (A + B + C): **≈ 13,997 DA**

Plus the parts not listed online: 18650 Li-ion battery, On/off switch, Capacitor 2200 µF 16 V, Capacitor 100 nF ("104"), Jumper wires male-female, Jumper wires male-male, Wire 22 AWG (0.5 mm²), Double-sided foam tape, M3 nylon standoffs + screws. Free delivery over 5000 DA.

D · Tools (if you don't have them)

✓	Tool	Why	dzduino price
☐	**Multimeter** Any digital multimeter	Required: check battery polarity and the 5 V rail	**12,000 DA** Fluke 101 (ask for a cheaper model) **In Stock**
☐	**Soldering iron + solder** 30–60 W, 0.8 mm solder	Motor wires and capacitors	**Not online: ask at the counter**
☐	**Small screwdriver set** Flat + Phillips	L298N screw terminals	**Not online: ask at the counter**
☐	**Wire stripper / cutter**		**Not online: ask at the counter**

E • Optional upgrades

✓	Part	Qty	Why	dzduino price
☐	**Ultrasonic sensor mounting bracket** For HC-SR04	1	Holds the sensor at the rear	**150 DA** Ultrasonic Sensor Mounting Bracket **In Stock**
☐	**SG90 micro servo** + SG90 bracket	1	Scanning radar (USE_SERVO). Needs its own 5 V supply	**150 DA** SG90 servo bracket (servo: ask) **In Stock**
☐	**FTDI USB-serial adapter** FT232RL / CP2102, 3.3 V / 5 V	1	Only if no ESP32-CAM-MB	**900 DA** FT232RL FTDI USB To TTL Serial Converter **Out of Stock online**
☐	**2.4 GHz IPEX antenna** For ESP32-CAM	1	Longer WiFi range	**Not online: ask at the counter**
☐	**Micro-USB cable (data)** For the ESP32-CAM-MB	1	Skip if you have one	**Not online: ask at the counter**

Check before you pay

Buzzer must be ACTIVE
Active = it beeps when you connect 5 V. Ask the seller to test it, or look for a sealed back with a sticker. A passive one stays silent on DC.

ESP32-CAM: AI-Thinker + OV2640
Check the camera ribbon is in the box. Buying it together with the MB programmer board avoids the FTDI wiring.

18650 batteries
Real cells weigh about 45 g. Very light cells with "9900 mAh" are fake and will make the ESP32-CAM reboot.

Chassis kit contents
If the kit already has the battery holder, switch or wires, don't buy them twice: tick them off in the list.

Why these quantities: 4 resistors 220 Ω (red LED A0, white reverse LED A2, 2 white headlights A4: one resistor each), 2200 µF at the ESP32-CAM, 1 kΩ + 2 kΩ divider (UNO D2 → ESP32 IO15), 4 × 100 nF (one per motor). About 18 male-female and 14 male-male jumpers are used; buy packs of 40 for spares. Prices on dzduino.com change: check them in the shop.